

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE CODE: CSA04
COURSE NAME: Operating Systems

COURSE OUTCOME COVERED

CO4: Optimize various file allocation strategies and disk scheduling algorithms to identify optimal methods for enhancing system performance.

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

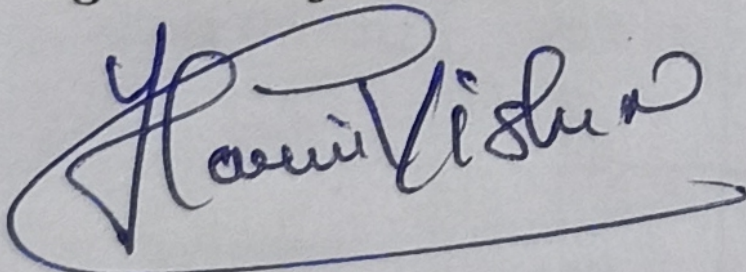
COMPARATIVE ANALYSIS

Code of Conduct:

I HARI VISHNU N / 192524319 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

COMPARATIVE ANALYSIS

S.No	Comparative Analysis Questions	Marks
1	Compare Contiguous and Linked file allocation methods in terms of access speed, memory utilization, and fragmentation. Identify which method is more suitable for sequential file access and justify your answer.	5
2	Differentiate between Linked and Indexed file allocation with respect to random access capability, storage overhead, and reliability. Explain which method is more efficient for large files.	5
3	Compare FCFS and SSTF disk scheduling algorithms based on seek time, fairness, and the possibility of starvation. Discuss which algorithm performs better under heavy disk load.	5
4	Analyze the differences between SCAN and C-SCAN disk scheduling algorithms in terms of head movement, response time, and uniformity of service. Which algorithm is more suitable for real-time systems and why?	5
5	Compare LOOK and C-LOOK disk scheduling algorithms with respect to efficiency, total head movement, and practical implementation. Explain which algorithm provides better performance and under what conditions.	5

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Test Case Handling	15%	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases

1) Contiguous vs Linked file Allocation :-

Feature	Contiguous	Linked
Access Speed	Very fast, supports direct access	Slower because links must be followed
Memory utilization	May waste space due to fragmentation	Better utilization of free blocks
Fragmentation	Suffers from external fragmentation	No external fragmentation
Sequential Access	Excellent	Good.
Random Access	Very good	Poor

Conclusion :- Contiguous allocation is more suitable for sequential file access because all the file blocks are stored consecutively, reducing seek time and providing faster access.

2) Linked vs Indexed file Allocation :-

Feature	Linked	Indexed
Random access	Poor	Good
Storage overhead	Pointer required in each block	Requires separate index blocks.
Reliability	A damaged pointer can break the chain	More reliable for accessing blocks.
Large files	Can be suitable but access become slower	More suitable
Access speed	Slower	Faster

Conclusion:- Indexed allocation is more efficient for large file because it supports direct/random access and provides quick access to any file block through the index block.

3) FCFS vs SSTF

Feature	FCFS	SSTF
Seek Time	usually higher	usually lower
Fairness	very high	Lower
Starvation	No starvation	Possible
Implementation	very simple	More complex
Heavy load	Performance decreases	Generally better

Conclusion:-

SSTF performs better under heavy disk load because it services the closest request first, reducing average seek time. However FCFS is fairer because every request is served in arrival order.

4) SCAN vs C-SCAN

Feature	SCAN	C-SCAN
Head Movement	Moves in both direction	Moves in one direction
Response Time	Good	More uniform
Service uniformity	less uniform	High uniform
Fairness	Good	Better
Real time suitability	Good	Better for predictable access

Conclusion :-

C-SCAN is more suitable for real time system when predictable and uniform response time are important. It provides more consistent waiting time because request are serviced in a circular manner.

5) Look vs C-Look

Feature	Look	C-Look
Efficiency	High	Very high
Head Mov.	Reverses at last request	Jumps from last request to first
Total Movement	Usually higher	Usually lower/more uniform
Implement	Simple	Slightly complex
Service	Both direction	only one.

Conclusion: C-look generally provide better performance when uniform service and reduced unnecessary head movement are required.